



Chap 11 and 12: Dictionaries and Tuples

CS 101- Programming
Fundamentals

Instructor: Neelma Bhatti

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Dictionaries

- Python dictionary is an ordered collection of items.
- Comprises of collection of indices (keys) and a collection of values.
- Each key is associated with a single value.
- The association of a key and a value is called a **key-value pair** or sometimes an **item**.

Dictionaries

- Creating a dictionary

```
Marks = {"Pfun":100, "DYHE":50, "FS":90}
```

```
Marks = dict({"Pfun":100, "DYHE":50, "FS":90})
```

- Accessing items:

```
Marks["Pfun"]
```

```
Marks.get("Pfun")
```

- Adding/modifying values

Dictionary as a collection of counters

- **Problem:** counting the number of times a character appears in a string
- **Possible approaches?**
- **Nested dictionaries**

Dictionary vs List

- **List:** When you just need a sequence of elements that you can access with indexing.
- **Dictionary:** If you need to *quickly* access an element mapped to a specific *unique* key, choose a dictionary.

Tuples overview

- An **immutable** sequence of values.
- Values can be of any type, indexed by integers.
- For single value tuple: **tupple_name = value ,**
- For multivalued tuples:
 - **tupple_name = comma separated values**
 - **tupple_name = (comma separated values)**

Tuples as return values

- Multiple values can be returned as tuple (comma separated)
- Example: *divmod* takes two arguments and returns a tuple of two values, quotient and remainder.

```
quot, rem = divmod( value1, value2)
```

```
return quot, rem
```